

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Objective type questions. (Compulsory Question)		12
1 Why is modulation necessary in a communication system?		1
a) To reduce signal power b) To prevent signal distortion		
c) To enable long-distance transmission d) To minimize interference		
2 Sampling theorem states that a continuous-time signal can be completely represented in its samples if the sampling frequency is:		1
a) Equal to the signal frequency b) Twice the highest signal frequency		
c) Half the highest signal frequency d) Independent of the signal frequency		
3 Multiplexing is used in communication systems to:		1
a) Reduce bandwidth requirements b) Combine multiple signals into one		
c) Enhance signal strength d) Improve signal quality		
4 What happens when the modulation index exceeds 1 in AM?		1
a) Improved signal quality b) Overmodulation, leading to distortion		
c) Reduced carrier power d) No effect on signal		
5 The bandwidth of an AM signal is equal to:		1
a) Twice the message signal frequency b) The carrier frequency		
c) Half the message signal frequency d) The message signal frequency		
6 Which type of AM has a full carrier and both sidebands?		1
a) DSB-FC b) DSB-SC c) SSB-SC d) VSB		
7 In PM, the instantaneous phase deviation is proportional to:		1
a) Integral of the message signal b) Amplitude of the message signal		
c) Frequency of the carrier signal d) Bandwidth of the carrier		
8 De-emphasis is applied at the receiver to:		1
a) Reduce high-frequency noise b) Amplify low-frequency signals		
c) Cancel pre-emphasis effects d) Equalize the carrier signal		
9 Image Frequency Rejection Ratio (IFRR) measures the receiver's ability to:		1
a) Reject noise in the signal b) Suppress image frequencies		
c) Amplify weak signals d) Enhance signal fidelity		
10 AGC helps in improving:		1
a) Fidelity of the receiver b) Selectivity of the receiver		
c) Stability of the signal strength d) Oscillator frequency stability		
11 Which of the following is a natural source of noise?		1
a) Thermal noise b) Cosmic noise c) Industrial noise d) Shot noise		
12 The envelope detector is commonly used for:		1
a) Frequency Modulated (FM) signals b) Phase Modulated (PM) signals		
c) Amplitude Modulated (AM) signals d) Suppressing noise		

Q. 2 Solve the following.		12
A) Differentiate between simplex, half-duplex, and full-duplex communication systems with examples.	CO 01, 07	6
B) Define Modulation, explain necessity of modulation in a communication system.	CO 02, 04	6
Q.3 Solve the following.		12
A) Explain DSBSC variant of AM. Derive the expression for the total transmitted power of a DSB-SC signal.	CO 02, 03	6
B) A VSB transmitter that transmits 20 % of the opposite sideband along with the desired sideband radiates 125 W when the modulation percentage is 70%. Calculate the total power required to transmit the same message using an AM transmitter.	CO 02, 04	6
Q. 4 Solve Any Two of the following.		12
A) Compare Amplitude Modulation (AM), Frequency Modulation (FM), and Phase Modulation (PM).	CO 05, 07	6
B) Define the terms : (a) Pre-emphasis, (b) De-emphasis, (c) Narrowband FM and (d) Wideband FM	CO 05	6
C) In an FM system, when the Audio Frequency (AF) is 700 Hz, and the AF voltage is 1.2 V, the deviation is 2.4 kHz. If the AF voltage is now increased by 3.6 V, what is new deviation? If the AF voltage is further raised to 7.2 V while the AF is dropped to 500 Hz, What is the deviation? Find modulation index in each case.	CO 05, 06	6
Q.5 Solve Any Two of the following.		12
A) Explain the performance characteristics of the receiver.	CO 04, 06	6
B) Describe the Tuned Radio Frequency (TRF) receiver architecture and its advantages and disadvantages.	CO 04	6
C) Explain double spotting with its cause and effect. Explain how it can be reduced.	CO 04	6
Q. 6 Solve Any Two of the following.		12
A) Explain operation of ratio detector. Also explain how amplitude limiting can be achieved using ratio detector.	CO 03, 06	6
B) Define Noise Figure, Noise Factor, and explain their relationship in evaluating the noise performance of electronic devices.	CO 06	6
C) Explain the operation of an Envelope Detector in AM demodulation, and Discuss its advantages and limitations.	CO 02, 04	6

*** End ***